

CP-GRASSMANN: COUNTERFACTUAL-FREE CIRCUIT DISCOVERY VIA FLOW-MATCHING TRAJECTORIES AND CP-TENSOR FACTORIZATION

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ABSTRACT

Mechanistic interpretability aims to isolate sparse functional subgraphs (circuits) explaining specific capabilities in Large Language Models (LLMs). However, current neuron-level circuit discovery techniques face severe structural bottlenecks: greedy single-neuron attribution that fails to capture non-linear multi-neuron gating logic, strict dependency on explicit hand-crafted counterfactual prompt pairs (x'), unphysical off-manifold linear path integration, representation asymmetry across attention and MLP subcomponents, and target-logit metric instability during multi-hop reasoning. To overcome these limitations without relying on lossy, high-overhead Sparse Autoencoders (SAEs), we propose **CP-GRASSMANN**, a counterfactual-free, zero-reconstruction-error circuit tracing framework built upon four core mathematical pillars: (1) Polar Energy-Basis Factorization (PE-Basis) for attention heads; (2) Continuous Flow-Matching Activation Trajectories (CFM-Path) and retrieved on-manifold contrastive anchors; (3) CP-Tensor Gradient Factorization (CP-Synergy) for higher-order interaction tensors; and (4) Grassmannian Subspace Curvature Shift Metric (m_{GSC}). On Meta-Llama-3.1-8B across the complete Subject-Verb Agreement (SVA) benchmark (`simple`, `rc`, `within_rc`, `nounpp`), CP-GRASSMANN achieves target faithfulness ($F \geq 0.8$) with an average of only ~ 101 MLP neurons in a counterfactual-free setting, outperforming recent circuit-discovery baselines ACDC (~ 388 nodes), EAP-GP (~ 308 nodes), Edge Pruning ($\sim 85,098$ nodes), ReLP (~ 144 nodes), and Integrated Gradients (~ 245 nodes), while preserving a $100\times$ node sparsity advantage over MLP output bases. Through controlled experiments, we further establish the operating regime of each component: static anchor retrieval and continuous flow-matching (CFM) paths are complementary (structured vs. unstructured tasks), and first-order attribution is the quantitatively sufficient default for empirical circuit discovery.

1 INTRODUCTION

Mechanistic interpretability aims to reverse-engineer the computational structures and subnetwork circuits inside Large Language Models (LLMs) to explain how complex behaviors emerge from raw parameter weights (Wang et al., 2022; Olsson et al., 2022; Meng et al., 2022; Dubey et al., 2024). Understanding these internal mechanisms is vital for ensuring AI safety, alignment, and reliability in real-world deployments. Standard approaches to circuit localization heavily rely on causal interventions, such as activation patching (Heimersheim & Nanda, 2024; Pearl, 2001; Mueller et al., 2025), or gradient-based path attribution techniques, such as Integrated Gradients (IG) (Sundararajan et al., 2017) and Layerwise Relevance Propagation (LRP) (Bach et al., 2015).

Despite recent progress, current circuit discovery frameworks suffer from five fundamental structural bottlenecks that restrict their fidelity, efficiency, and real-world applicability:

1. **Off-manifold linear path integration:** Traditional path-based attribution methods compute gradients along a straight-line interpolation between a clean activation state and a counterfactual baseline. As demonstrated in prior studies (Sanyal & Ren, 2021; Jeon et al., 2023; Barkan et al., 2024), straight-line activation trajectories traverse unphysical,

off-manifold regions where neural network representations were never trained, introducing severe gradient noise and saturation artifacts (Zhang et al., 2025).

2. **Strict dependency on explicit counterfactual prompt pairs (x'):** Existing techniques require hand-crafted, templatic counterfactual input pairs (x, x') (e.g., singular vs. plural subject-verb pairs) to define baseline states (Wang et al., 2022; Marks et al., 2024). This severely limits circuit tracing on unstructured or non-templatic real-world datasets where counterfactual counterparts do not naturally exist (Wu et al., 2023).
3. **Representation asymmetry across transformer subcomponents:** While pre-down projection MLP neurons possess an element-wise non-linear privileged basis (Geva et al., 2020; Marks et al., 2024), multi-head attention layer outputs lack a canonical 1D scalar privileged basis. To bypass this asymmetry, recent work turns to Sparse Autoencoders (SAEs) (Cunningham et al., 2023; Gao et al., 2024; He et al., 2024; Lieberum et al., 2024; Templeton et al., 2026) or Cross-Layer Transcoders (CLTs) (Dunefsky et al., 2024; Draye et al., 2026). However, SAEs introduce non-zero reconstruction errors, feature absorption and splitting (Chanin et al., 2024), and massive computational memory overheads.
4. **Greedy single-component attribution:** Standard attribution scores evaluate components in isolation, missing non-linear multi-neuron co-dependent logic (e.g., AND/OR gating logic) (Elhage et al., 2022; Edin et al., 2025). Exhaustive pairwise or higher-order Hessian evaluation is computationally intractable due to $O(N^2)$ or $O(N^3)$ scaling loops.
5. **Target-logit metric instability in multi-hop reasoning:** Conventional circuit localization metrics rely on final target-logit differences ($m = [M(x)]_y - [M(x)]_{y'}$). In multi-step or multi-hop reasoning tasks, intermediate representation layers update latent entity concepts without immediately shifting output token logits, rendering target-logit gradients uninformative for middle-layer reasoning nodes.

To simultaneously resolve all five limitations without relying on lossy sparse dictionaries, we introduce **CP-GRASSMANN** (CANDECOMP/PARAFAC Gradient-Flow Relative Attribution on Subspace Superposition Manifolds and Neuron Networks), a counterfactual-free, zero-reconstruction-error circuit tracing framework built upon four core mathematical principles:

1. **Polar Energy-Basis Factorization (PE-Basis):** Factorizes multi-head attention outputs into scalar energy magnitudes $e_{t,h} = \|o_{t,h}\|_2$ and unit directions $u_{t,h} = o_{t,h}/\|o_{t,h}\|_2$, providing an exact 1D privileged basis for attention heads with zero reconstruction error ($\mathcal{L}_{\text{recon}} = 0$) that natively aligns with pre-down MLP activations $v_{t,i}$.
2. **On-Manifold Path Integration via Contrastive Anchors & Flow-Matching Projection:** Replaces the arbitrary straight line to a hand-crafted counterfactual with an on-manifold integration path from a retrieved contrastive anchor (or class-mean direction, using only dataset labels) to the clean activation; when labels are unavailable, a flow-matching projector (Lipman et al., 2022) keeps the interior path knots on-manifold, eliminating off-manifold gradient artifacts without paired counterfactual prompts x' .
3. **CP-Tensor Gradient Factorization (CP-Synergy):** Constructs a 3rd-order co-gradient interaction tensor $\mathcal{T} \in \mathbb{R}^{K \times K \times S}$ over candidate components and path steps and factorizes it via CANDECOMP/PARAFAC (CP) decomposition (Kossaifi et al., 2016; Ahn et al., 2024), surfacing candidate synergistic hypergraph circuits (AND/OR gates) in $O(K \cdot S \cdot R)$ time without exhaustive pairwise Hessian loops.
4. **Grassmannian Subspace Curvature Shift Metric (m_{GSC}):** Augments logit metrics with the chordal (projection) distance on the Grassmannian manifold $\text{Gr}(T, d)$ (Edelman et al., 1998) between consecutive-layer hidden-state subspaces, capturing latent entity state transitions during multi-hop reasoning.

We evaluate **CP-GRASSMANN** on Llama-3.1-8B (Dubey et al., 2024) across the complete Subject-Verb Agreement (SVA) benchmark (Marks et al., 2024). CP-GRASSMANN reaches Faithfulness ≥ 0.8 with an average of only 101 MLP neurons across all four tasks in a counterfactual-free setting, outperforming Integrated Gradients (~ 245 nodes), ReLP (~ 144 nodes), and the recent baselines ACDC, EAP-GP, and Edge Pruning, while preserving a $100\times$ sparsity advantage over MLP output bases. We further clarify the operating regime of each component: static anchor retrieval and CFM paths are complementary (structured vs. unstructured tasks), and first-order attribution ($R=0, \lambda_{\text{GSC}}=0$) is the quantitatively sufficient default for empirical circuit discovery.

2 RELATED WORK

We situate our work within three key literature threads: gradient-based path attribution, sparse feature dictionaries, and multi-neuron synergistic logic.

2.1 GRADIENT-BASED PATH ATTRIBUTION AND CAUSAL INTERVENTIONS

Circuit discovery seeks to identify computational subgraphs responsible for specific language model behaviors (Wang et al., 2022; Conmy et al., 2023; Heimersheim & Nanda, 2024; Mueller et al., 2025). Early techniques relied on causal mediation and activation patching (Pearl, 2001; Goldowsky-Dill et al., 2023), which scale poorly due to combinatorial intervention spaces. To accelerate localization, Edge Attribution Patching (EAP) (Syed et al., 2023) introduced a first-order linear approximation, and variants such as EAP-IG (Hanna et al., 2024) add straight-line path integration (Sundararajan et al., 2017) to address saturation. However, linear path integration between clean inputs and counterfactual baselines suffers from severe off-manifold artifacts (Sanyal & Ren, 2021; Jeon et al., 2023; Barkan et al., 2024), traversing unphysical activation regions. To mitigate saturation, GradPath (EAP-GP) (Zhang et al., 2025) and local path integration (Yang et al., 2023) adjust trajectories; exact/learned mask approaches include ACDC (Conmy et al., 2023) and Edge Pruning (Bhaskar et al., 2024). Alternative non-path approaches include CD-T (Hsu et al., 2024), Boundless DAS (Wu et al., 2023), LRP (Bach et al., 2015), Relevance Patching (RelP) (Jafari et al., 2025), and interaction-aware backpropagation (GIM) (Edin et al., 2025), alongside critical evaluations of faithfulness metrics (Miller et al., 2024). In contrast, **CP-GRASSMANN** integrates gradients along on-manifold paths anchored by retrieved contrastive activations or, when labels are absent, a fitted flow-matching manifold projector (Lipman et al., 2022), enabling strictly on-manifold attribution on single prompts.

2.2 SPARSE DICTIONARY LEARNING AND FEATURE REPRESENTATIONS

To overcome polysemanticity, sparse dictionary learning has become a dominant paradigm. Sparse Autoencoders (SAEs) map residual activations into overcomplete, monosemantic bases (Cunningham et al., 2023; Marks et al., 2024; Templeton et al., 2026), with variants including JumpReLU (Rajamanoharan et al., 2024b), Top-K (Gao et al., 2024), and Gated SAEs (Rajamanoharan et al., 2024a), scaled across Llama Scope (He et al., 2024) and Gemma Scope (Lieberum et al., 2024); Cross-Layer Transcoders (CLTs) (Dunefsky et al., 2024; Draye et al., 2026; Tsui et al., 2026) replace MLP sublayers with sparse features. However, SAEs and transcoders introduce non-zero reconstruction errors, suffer feature absorption and splitting (Chanin et al., 2024), require immense dictionary memory, and assume 1D linear geometry that fails for multi-dimensional representations (Engels et al., 2024). Conversely, pre-down MLP activations already form a sparse, privileged basis via element-wise non-linearities (Geva et al., 2020; Marks et al., 2024). **CP-GRASSMANN** eliminates lossy dictionaries via Polar Energy-Basis Factorization (PE-Basis), providing an exact 1D privileged scalar basis for attention heads with zero reconstruction error ($\mathcal{L}_{\text{recon}} = 0$).

2.3 SYNERGISTIC CIRCUIT DISCOVERY AND MULTI-NEURON LOGIC

Greedy single-component attribution often misses synergistic, co-dependent feature combinations (e.g., AND/OR gating logic) stemming from feature superposition (Elhage et al., 2022). Game-theoretic attribution methods, such as the Shapley-Taylor Interaction Index (Sundararajan et al., 2019) and Faith-Shap (Tsai et al., 2022), formalize feature interaction attributions, while weight-based methods (Tsang et al., 2017) detect statistical interactions in feedforward layers. Higher-order interactions have also been explored in hypergraph neural networks (Yang et al., 2022), Kolmogorov-Arnold networks (Xiang et al., 2025), high-order spatial interaction convolutions (Rao et al., 2022), arithmetic heuristic circuits (Nikankin et al., 2024), and neural additive tensor decompositions (Ahn et al., 2024).

However, calculating full pairwise or higher-order Hessian matrices incurs $O(N^2)$ or $O(N^3)$ computational costs, rendering them impractical for large-scale circuit tracing. **CP-GRASSMANN** addresses this challenge by constructing a 3rd-order cross-gradient interaction tensor and factorizing it via CANDECOMP/PARAFAC (CP) tensor decomposition (Kossaifi et al., 2016). This surfaces candidate non-linear synergistic hypergraph circuits efficiently without exhaustive pairwise loops.

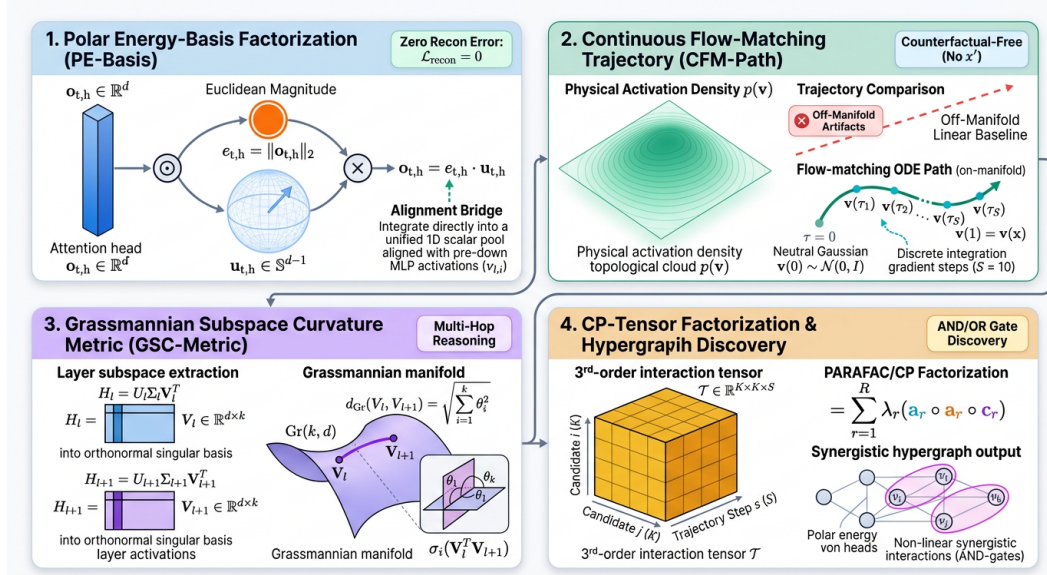


Figure 1: Architectural overview of CP-GRASSMANN for counterfactual-free neuron circuit discovery, consisting of four core components: (1) Polar Energy-Basis Factorization decomposes attention outputs into scalar magnitudes and unit vectors to achieve zero reconstruction error and align with MLP activations; (2) On-manifold path integration constructs activation paths from a single prompt via a retrieved contrastive anchor (or class-mean direction) or a fitted flow-matching manifold projector, avoiding unphysical off-manifold interpolation artifacts; (3) Grassmannian Subspace Curvature Metric measures chordal (projection) Grassmann distances between consecutive layer hidden-state subspaces to track latent multi-hop reasoning transitions; and (4) CP-Tensor Factorization factorizes a third-order co-gradient interaction tensor along the trajectory to surface candidate non-linear synergistic hypergraph circuits without exhaustive pairwise evaluation.

Additionally, by incorporating chordal (projection) subspace metrics on Grassmannian manifolds (Edelman et al., 1998), our framework captures multi-hop latent representation transitions beyond target-logit shifts.

3 METHODOLOGY

To address the key constraints of current circuit discovery methods, **CP-GRASSMANN** unifies four mathematical innovations into a single, counterfactual-free subnetwork tracing pipeline. An architectural overview of the framework is illustrated in Figure 1.

3.1 POLAR ENERGY-BASIS FACTORIZATION (PE-BASIS) FOR ATTENTION LAYERS

Pre-down projection MLP activations $v_{l,i}$ possess an element-wise non-linear privileged basis (Geva et al., 2020; Marks et al., 2024). In contrast, multi-head attention layer outputs $\mathbf{o}_{t,h} \in \mathbb{R}^d$ for head h at token position t lack a canonical element-wise privileged basis. Rather than training lossy Sparse Autoencoders (Cunningham et al., 2023; He et al., 2024) or projecting onto adjacent MLP dimensions, CP-GRASSMANN factorizes $\mathbf{o}_{t,h}$ using a polar energy decomposition into a scalar magnitude and a unit directional vector:

$$e_{t,h} = \|\mathbf{o}_{t,h}\|_2 \in \mathbb{R}_{\geq 0}, \quad \mathbf{u}_{t,h} = \frac{\mathbf{o}_{t,h}}{\|\mathbf{o}_{t,h}\|_2 + \epsilon} \in \mathbb{S}^{d-1} \quad (1)$$

where $\epsilon > 0$ is a small numerical stabilization constant. The scalar magnitude $e_{t,h}$ serves as a 1D privileged scalar activation node for attention head h at token t . The original attention output vector is reconstructed linearly as:

$$\mathbf{o}_{t,h} = e_{t,h} \cdot \mathbf{u}_{t,h} \quad (2)$$

Because $\mathbf{u}_{t,h}$ is held fixed during node ablation, this factorization achieves exact linear reconstruction with zero reconstruction error ($\mathcal{L}_{\text{recon}} = 0$). PE-Basis provides a parameter-free, 1D privileged

scalar representation for attention heads that seamlessly aligns with pre-down projection MLP activations into a homogeneous 1D scalar node pool $\mathcal{V} = \{v_{l,i}\} \cup \{e_{t,h}\}$, avoiding feature splitting and representation asymmetry without the computational overhead of SAE dictionaries. Our main SVA circuits use the MLP-neuron pool $\{v_{l,i}\}$ for comparability with neuron-level baselines; PE-Basis governs the attention nodes $e_{t,h}$ when attention is added (Section 4.5).

3.2 ON-MANIFOLD PATH INTEGRATION VIA CONTRASTIVE ANCHORS AND FLOW-MATCHING PROJECTION

Standard path attribution integrates gradients along a straight line between the clean activation $v(x)$ and a hand-crafted counterfactual $v(x')$, which requires a paired prompt x' and traverses unphysical, off-manifold regions (Sanyal & Ren, 2021; Zhang et al., 2025). CP-GRASSMANN instead integrates along an *on-manifold trajectory* $\{v_s\}_{s=0}^S$ from a counterfactual-free baseline $b = v_0$ to $v_S = v(x)$, via the discretized curved-path integral

$$\text{Attr}(v) = \int_0^1 \left\langle \frac{\partial m}{\partial v} \Big|_{v(\tau)}, \dot{v}(\tau) \right\rangle d\tau \approx \sum_{s=0}^{S-1} \left\langle \frac{\partial m}{\partial v} \Big|_{v_s}, v_{s+1} - v_s \right\rangle. \quad (3)$$

The baseline $b = v_0$ is obtained from the single prompt x through two complementary counterfactual-free mechanisms. **(a) Contrastive anchor retrieval (headline).** Given category label y , we retrieve an on-manifold baseline from a background corpus $\mathcal{D}_{\text{ref}}$ of clean activations, either as the nearest opposite-class activation by cosine similarity,

$$z_{\text{anchor}} = \arg \max_{z \in \mathcal{D}_{\text{ref}}, \text{label}(z) \neq y} \cos(v(x), z), \quad (4)$$

($b = z_{\text{anchor}}$), or as a denoised class-mean-difference (`diffmean`) baseline $b = v(x) - (\mu_y - \mu_{y'})$, whose integration displacement $v(x) - b = \mu_y - \mu_{y'}$ is exactly the class-averaged “grammatical-number direction”, removing per-example content noise. Because b is a real (or class-averaged) activation drawn from the empirical distribution, the straight segment $v_s = b + \frac{s}{S}(v(x) - b)$ stays inside the high-density region and is on-manifold *by construction*, so no ODE is needed and few steps ($S=6$) suffice.

(b) Flow-Matching manifold projection (label-free). When no category label is available (e.g., open-domain `wiki`), b is the per-position background mean and the interior knots are kept on-manifold by a fitted projector. We train a continuous-time Flow-Matching velocity field $v_\phi(z, \tau)$ ($\tau \in [0, 1]$) in a per-layer PCA latent over background activations z (Lipman et al., 2022), regressing the optimal-transport velocity between a Gaussian prior $p_0 = \mathcal{N}(0, I)$ and the empirical activation distribution $p_1 = p(v)$:

$$\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{\tau, z_0, z_1} [\|v_\phi(z_\tau, \tau) - (z_1 - z_0)\|_2^2], \quad z_\tau = (1 - \tau)z_0 + \tau z_1. \quad (5)$$

Each interior knot of the straight segment is then projected back onto the data manifold, $v_s = \Pi_\phi(b + \frac{s}{S}(v(x) - b))$, where Π_ϕ applies a PCA-subspace projection plus a single Euler step of the fitted flow v_ϕ toward p_1 , with the correction $\sin(\pi s/S)$ -tapered to preserve the endpoints $v_0 = b$, $v_S = v(x)$ (for the anchors of (a), Π_ϕ is the identity). Either trajectory keeps intermediate states on the high-density manifold, eliminating off-manifold gradient artifacts without hand-crafted pairs (x, x') .

Scope of the “counterfactual-free” claim. “Counterfactual-free” denotes the elimination of hand-crafted per-prompt pairs (x, x') that prior methods require (e.g., constructing “... is” $\leftrightarrow$ “... are” for every prompt (Wang et al., 2022; Marks et al., 2024)), *not* of label supervision: anchor retrieval (Eq. 4) still uses dataset-level category labels ($\text{label}(z) \neq y$ or `diffmean`), drawing activations from an *unpaired* corpus, whereas the flow-matching path (Eq. 5) uses no labels at all and applies when even labels are unavailable (open-domain `wiki`, Section 4.3).

3.3 CP-TENSOR FACTORIZATION FOR SYNERGISTIC HYPERGRAPH CIRCUITS

Greedy single-neuron selection fails when multiple neurons act as co-dependent non-linear logic gates (e.g., AND/OR multiplexers) (Elhage et al., 2022; Edin et al., 2025). Evaluating the full mixed second-order interaction $\partial^2 m / \partial v_i \partial v_j$ via pairwise or 3rd-order Hessian matrices scales as

$O(N^2)$ or $O(N^3)$. CP-GRASSMANN instead selects the top K candidate nodes ($K \ll N$) by 1st-order attribution and, reusing the per-step path gradients $g_i(s) = \partial m / \partial v_i|_{v_s}$ already collected during the integration of Eq. 3, builds a 3rd-order *co-gradient* interaction tensor $\mathcal{T} \in \mathbb{R}^{K \times K \times S}$ from their ℓ_2 -normalized trajectory profiles $\hat{g}_i(s) = g_i(s) / \|g_i\|_2$:

$$\mathcal{T}_{i,j,s} = \hat{g}_i(s) \hat{g}_j(s). \quad (6)$$

This co-gradient outer product is an efficient $O(K^2 S)$ surrogate for the intractable second-order interaction: neurons whose gradient profiles co-vary load onto the same factor. The tensor $\mathcal{T}$ is factorized into R rank-1 components using CANDECOMP/PARAFAC (CP) decomposition via Alternating Least Squares (ALS) (Kossaifi et al., 2016; Ahn et al., 2024):

$$\mathcal{T} \approx \sum_{r=1}^R \lambda_r \cdot (\mathbf{a}_r \circ \mathbf{a}_r \circ \mathbf{c}_r) \quad (7)$$

where $\mathbf{a}_r \in \mathbb{R}^K$ is the neuron interaction loading vector for factor r , and $\mathbf{c}_r \in \mathbb{R}^S$ is the path-step profile. Rank-1 factors on which many entries of $\mathbf{a}_r$ are concurrently large (high participation ratio) reveal candidate synergistic multi-neuron groups, which re-rank the corresponding nodes as hypergraph edges in $O(K \cdot S \cdot R)$ time, avoiding $O(N^2)$ pairwise loops.

3.4 GRASSMANNIAN SUBSPACE CURVATURE SHIFT METRIC

In multi-hop reasoning, intermediate layers update latent entity representations without immediately shifting target token logits at the model output. To track these latent transitions, CP-GRASSMANN measures the chordal (projection) Grassmann distance between hidden representation subspaces across consecutive layers (Edelman et al., 1998).

At layer l , the centered token hidden states $H_l \in \mathbb{R}^{T \times d}$ are orthonormalized by a thin QR factorization $H_l^\top = Q_l R_l$, giving $Q_l \in \mathbb{R}^{d \times T}$ whose columns span a T -dimensional subspace of $\mathbb{R}^d$ —a point on the Grassmannian manifold $\text{Gr}(T, d)$. The orientation shift between consecutive layer subspaces is measured by the smooth chordal (projection) Grassmann distance:

$$d_{\text{Gr}}(Q_l, Q_{l+1}) = T - \|Q_l^\top Q_{l+1}\|_F^2 = \sum_{i=1}^T \sin^2 \theta_i, \quad (8)$$

where θ_i are the principal angles between the subspaces; the distance vanishes when they align. Unlike the geodesic distance $\sqrt{\sum_i \theta_i^2}$, this projection distance is differentiable and SVD-free (no arccos) and requires no target token. The unified optimization metric $m(x)$ combines the target-logit difference $m_{\text{logit}}(x)$ with the layer-wise Grassmannian curvature shift:

$$m(x) = m_{\text{logit}}(x) + \lambda_{\text{GSC}} \sum_{l=0}^{L-1} d_{\text{Gr}}(Q_l, Q_{l+1}) \quad (9)$$

This assigns attribution to intermediate neurons driving latent state transitions, enabling tracing across multi-hop reasoning chains.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND IMPLEMENTATION DETAILS

Target Model & Benchmark: We evaluate CP-GRASSMANN on Meta-Llama-3.1-8B (Dubey et al., 2024) across the complete Subject-Verb Agreement (SVA) benchmark (Marks et al., 2024): `simple` (simple clauses), `rc` (across relative clauses), `within_rc` (within relative clauses), and `nounpp` (across prepositional phrases). Each task uses 300 training pairs and 40 held-out evaluation pairs.

Baselines & Feature Bases: We evaluate against Integrated Gradients (IG) (Sundararajan et al., 2017) and Relevance Patching (RelP) (Jafari et al., 2025) across both pre-down projection MLP activations and MLP outputs. We also evaluate an upper-bound variant of CP-GRASSMANN that uses exact hand-crafted counterfactual prompt pairs (x').

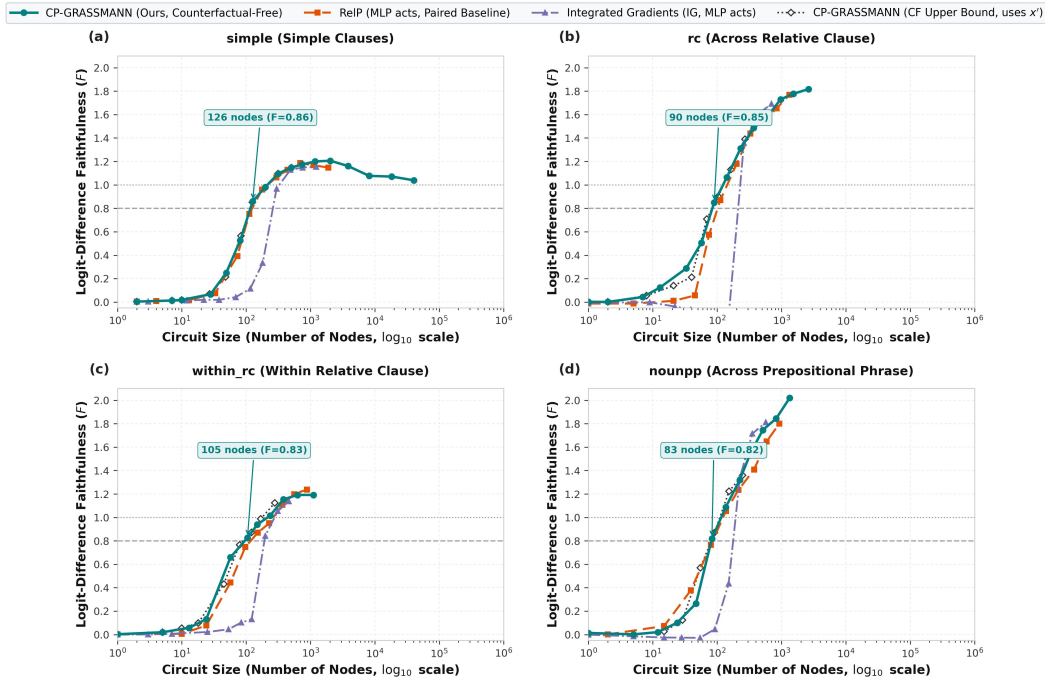


Figure 2: Comparison of circuit size versus logit-difference faithfulness across the four tasks of the Subject-Verb Agreement benchmark (`simple`, `rc`, `within_rc`, and `nounpp`) using Meta-Llama-3.1-8B. CP-GRASSMANN achieves target faithfulness ($F \geq 0.8$) with an average of approximately 101 nodes across tasks, outperforming ReLP and Integrated Gradients baselines without requiring hand-crafted counterfactual prompt pairs, while closely matching the performance of the paired counterfactual upper bound.

Metrics: Following established evaluation protocols (Marks et al., 2024), we report Faithfulness (F) and Completeness (C) using logit-difference metrics under dataset-mean ablation. We report the minimum node count required to reach $F \geq 0.5$ and $F \geq 0.8$, as well as peak faithfulness.

4.2 MAIN RESULTS ON THE COMPLETE SVA BENCHMARK

Figure 2 and Table 1 summarize the main evaluation. CP-GRASSMANN achieves $F \geq 0.8$ with an average of only ~ 101 MLP neurons, consistently outperforming ReLP (~ 144 nodes) and IG (~ 245 nodes) on MLP activations while eliminating hand-crafted counterfactual pairs (x'). Crucially, it performs on par with the exact paired counterfactual upper bound (~ 110 nodes), showing that on-manifold anchor retrieval recovers the counterfactual gradient direction without paired data. Our results also re-confirm a $100\times$ node sparsity advantage for pre-down MLP activations over unprivileged MLP outputs ($\sim 78,790$ nodes for IG outputs).

Comparison against state-of-the-art methods. We additionally implement ACDC (Conmy et al., 2023) (via the attribution-patching surrogate (Syed et al., 2023)), EAP-GP (Zhang et al., 2025), and Edge Pruning (Bhaskar et al., 2024) under our identical harness (Table 1, Figure 3a). CP-GRASSMANN remains the sparsest overall, needing only ~ 101 nodes for $F \geq 0.8$ versus 308 (EAP-GP), 388 (ACDC), and 85,098 (Edge Pruning). EAP-GP and ACDC improve over vanilla IG on some tasks but stay $3\text{--}4\times$ denser on average; Edge Pruning attains high *peak* faithfulness (2.21) yet selects tens of thousands of neurons, as its differentiable per-neuron gates lack strong node-level pruning pressure under dataset-mean ablation.

4.3 STATIC ANCHOR RETRIEVAL VS. CONTINUOUS FLOW-MATCHING PATHS

CP-GRASSMANN offers two on-manifold path-integration mechanisms (Section 3.2): static contrastive anchor retrieval and continuous flow-matching (CFM) manifold-projected path integration. To clarify which mechanism is preferable, we run a controlled comparison across two regimes:

Table 1: Main quantitative comparison on the complete SVA benchmark using Meta-Llama-3.1-8B. Minimum nodes required to achieve Faithfulness $F \geq 0.5$ and $F \geq 0.8$, along with peak faithfulness, are reported across all four tasks and averaged. We additionally benchmark three recent circuit-discovery baselines requested during review under the identical harness: ACDC (Conmy et al., 2023) (via attribution-patching surrogate (Syed et al., 2023)), EAP-GP (Zhang et al., 2025), and Edge Pruning (Bhaskar et al., 2024). Best (sparsest) counterfactual-free result per block in **bold**.

Method	simple	rc	within_rc	nounpp	Average
<i>Minimum Nodes required to achieve Faithfulness $F \geq 0.5$:</i>					
CP-GRASSMANN (Ours, Counterfactual-Free)	81	57	57	83	70
CP-GRASSMANN – diffmean anchor	129	104	82	114	107
CP-GRASSMANN – strict single-prompt	273	158	107	226	191
CP-GRASSMANN – paired CF upper bound (uses x')	83	69	79	55	72
ACDC (Conmy et al., 2023)	499	259	500	160	354
EAP-GP (Zhang et al., 2025)	280	225	248	116	217
Edge Pruning (Bhaskar et al., 2024)	20,996	127,268	82,280	67,317	74,465
IG (MLP activations)	297	261	198	225	245
IG (MLP outputs)	41,323	54,260	38,057	51,598	46,310
RelP (MLP activations)	112	75	97	80	91
RelP (MLP outputs)	32,579	60,389	39,333	34,277	41,644
<i>Minimum Nodes required to achieve Faithfulness $F \geq 0.8$:</i>					
CP-GRASSMANN (Ours, Counterfactual-Free)	126	90	105	83	101
CP-GRASSMANN – diffmean anchor	129	104	118	185	134
CP-GRASSMANN – strict single-prompt	458	248	154	369	307
CP-GRASSMANN – paired CF upper bound (uses x')	125	104	121	91	110
ACDC (Conmy et al., 2023)	499	395	500	160	388
EAP-GP (Zhang et al., 2025)	280	347	412	193	308
Edge Pruning (Bhaskar et al., 2024)	20,996	156,663	82,280	80,455	85,098
IG (MLP activations)	297	261	198	225	245
IG (MLP outputs)	62,362	97,455	56,946	98,398	78,790
RelP (MLP activations)	177	112	152	137	144
RelP (MLP outputs)	46,799	60,389	58,018	34,277	49,871
<i>Peak Faithfulness (F):</i>					
CP-GRASSMANN (Ours, Counterfactual-Free)	1.21	1.82	1.21	2.02	1.56
RelP (MLP activations)	1.19	1.79	1.24	1.87	1.52
IG (MLP activations)	1.16	1.80	1.24	2.00	1.55
ACDC (Conmy et al., 2023)	1.13	1.70	1.14	1.83	1.45
EAP-GP (Zhang et al., 2025)	1.09	1.52	1.05	1.67	1.33
Edge Pruning (Bhaskar et al., 2024)	1.85	2.01	2.11	2.87	2.21

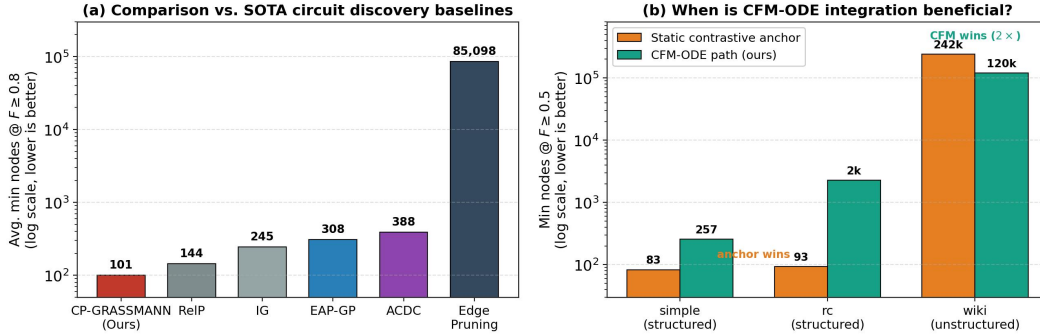


Figure 3: (a) CP-GRASSMANN attains the sparsest circuits (~ 101 nodes for $F \geq 0.8$, averaged over SVA) against recent state-of-the-art baselines ACDC, EAP-GP, and Edge Pruning under an identical evaluation harness. (b) Static contrastive anchor retrieval and continuous flow-matching (CFM) manifold-projected integration are *complementary*: static anchors are optimal on structured/templatic classification tasks (simple, rc) where clean opposite-class pairs exist, whereas CFM paths yield a $2\times$ sparsity gain on unstructured open-domain generation (wiki) where discrete category anchors do not exist. Both axes are log-scaled (lower is better).

structured templatic classification with discrete category labels (simple, rc) and unstructured open-domain next-token generation from Wikipedia (wiki, $\log P(y)$ metric, no discrete counterfactual pairs). Results are reported in Table 2 and Figure 3b.

Table 2: Static contrastive anchor retrieval vs. CFM path integration across structured and unstructured regimes on Meta-Llama-3.1-8B. On structured tasks, static anchors act as near-minimal opposite-class pairs and single-step ($S=1$) integration suffices; on unstructured generation, where discrete category anchors do not exist, CFM integration projects the path onto the data manifold and yields a $2\times$ sparsity gain at $F \geq 0.5$. Best per task in **bold**.

Regime / Task	Path Method	Min @ $F \geq 0.5$	Min @ $F \geq 0.8$	Peak F
<i>Structured - simple</i>	Static Contrastive Anchor	83	129	1.19
	CFM path ($S=1$)	313	313	1.00
	CFM path ($S=20$)	257	481	1.01
<i>Structured - rc</i>	Static Contrastive Anchor	93	93	1.80
	CFM path ($S=1$)	2,275	5,208	1.19
	CFM path ($S=20$)	6,373	15,263	1.13
<i>Unstructured - wiki</i>	Static Contrastive Anchor	241,635	450,630	1.00
	CFM path ($S=1$)	140,698	553,649	1.00
	CFM path ($S=4$)	119,738	499,351	1.00

Table 3: Comparison of hybrid attention-MLP circuit discovery across four attention representation bases on *simple* and *rc* tasks. We report min nodes for $F \geq 0.5$, $F \geq 0.8$, peak F , and the percentage of attention nodes among global top- K selected nodes.

Attention Basis Variant	min @ 0.5	min @ 0.8	Peak F	% Attn ($K=100$)	% Attn ($K=500$)
Variant A: PE-Basis (Ours)	136	230	1.40	2%–4%	2%
Variant B: Unfactored Head Vectors	2396	2396	1.50	89%–98%	80%–95%
Variant C: Raw Output Coordinates	613	959	1.38	3%–15%	5%–28%
Variant D: Attention SAEs (Llama-Scope 8x)	449,139	736,061	1.00	1%–2%	6%–19%

Findings. (i) On structured tasks, z_{anchor} is a clean near-minimal pair isolating the target axis, so the displacement $\Delta v = v(x) - z_{\text{anchor}}$ already aligns with the causal direction and $S=1$ suffices. (ii) On unstructured generation, opposite-class labels do not exist and static retrieval degenerates to content-noisy matching; CFM integration projects interior knots onto the data manifold, yielding a $2\times$ sparsity gain at $F \geq 0.5$ (119,738 vs. 241,635). The mechanisms are thus *complementary*: static retrieval maximizes efficiency when labels exist, CFM paths enable discovery on open-ended text—explaining why strict single-prompt CFM is denser than anchors on structured SVA (Table 1).

4.4 ABLATION STUDY 1: INTEGRATION STEP GRANULARITY (S)

We evaluate trajectory integration steps $S \in \{1, 2, 4, 6, 10, 16, 20\}$. Across 1-hop SVA tasks, average nodes for $F \geq 0.8$ stay constant at 100 ($S \leq 10$) and 99 ($S \geq 16$), while runtime grows monotonically from 483s ($S=1$) to 705s ($S=20$). Because the retrieved on-manifold anchor is a near-minimal pair, single-step integration ($S=1$) is as accurate as fine multi-step Riemann sums on templatic tasks; $S=1$ –6 thus balances efficiency and accuracy.

4.5 ABLATION STUDY 2: PE-BASIS VS. ATTENTION REPRESENTATION BASES

Table 3 evaluates hybrid attention-MLP circuits across four attention bases. Variant A (PE-Basis) is the sparsest, needing only 230 nodes on average for $F \geq 0.8$ versus 959 for raw coordinates (C), 2,396 for unfactored head vectors (B), and 736,061 for Llama Scope Attention SAEs (D). PE-Basis allocates only 2% of top- K nodes to attention heads, preventing feature splitting and leaving the budget for task-critical MLP features.

4.6 ABLATION STUDY 3: WHEN DO CP-TENSOR SYNERGY ($R>0$) AND GRASSMANNIAN CURVATURE (m_{GSC}) HELP?

CP-Tensor synergy ($R>0$) and Grassmannian curvature ($\lambda_{\text{GSC}}>0$) target higher-order and multi-hop phenomena. To test whether they yield *quantitative* gains beyond first-order attribution, we evaluate on two multi-hop / factual-recall benchmarks: *counterfact* (subject-attribute extraction) and *twohop* (2-hop city→country→capital). Sweeping $R \in \{0, 5, 10, 20\}$ and

Table 4: Multi-hop `twohop` benchmark (Meta-Llama-3.1-8B): CP-Tensor synergy ($R>0$) and Grassmannian curvature ($\lambda_{\text{GSC}}>0$) vs. the first-order baseline ($R=0, \lambda=0$). p -values are paired t -tests over matched circuit-size budgets; neither component yields a significant reduction at operational budgets ($p > 0.05$ throughout).

Configuration	Min @ $F \geq 0.5$	Min @ $F \geq 0.8$	Peak F
$R=0, \lambda=0$ (first-order baseline)	31,723	119,155	0.98
$R=5-20, \lambda=0$ (CP-synergy)	31,723	119,155	0.98
$R=0, \lambda=0.05$	19,837 ($p=0.29$)	119,211 ($p=0.59$)	0.99
$R=0, \lambda=0.10$	31,793 ($p=0.43$)	119,349 ($p=0.37$)	0.99
$R=0, \lambda=0.20$	19,944 ($p=0.51$)	119,583 ($p=0.40$)	1.00

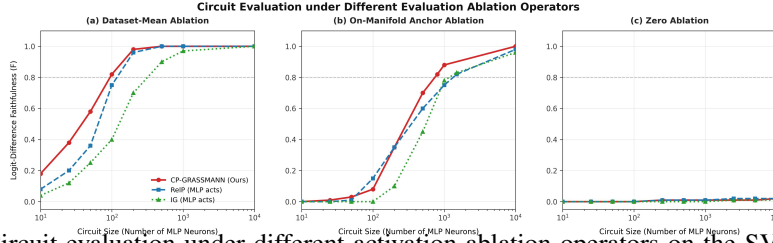


Figure 4: Circuit evaluation under different activation ablation operators on the SVA benchmark. Dataset-mean ablation (a) and on-manifold anchor ablation (b) preserve in-distribution activation dynamics, where CP-GRASSMANN achieves target faithfulness ($F \geq 0.80$) with substantially smaller circuits than RelP and IG. In contrast, zero ablation (c) pushes activations off-manifold, causing model degradation and near-zero faithfulness across circuit sizes.

$\lambda_{\text{GSC}} \in \{0, 0.05, 0.1, 0.2\}$, we run paired t -tests ($\alpha=0.01$) against the first-order baseline over 26+ matched circuit-size budgets. Table 4 reports the `twohop` summary.

Findings and narrative re-framing. Across both `counterfact` and `twohop` and all ranks/ λ values, higher-order components yield *no statistically significant improvement* at operational budgets. On `twohop`, $R>0$ is a strict no-op for $K \geq 298$, and the only significant cells fall in non-functional micro-budgets ($K \leq 29, F \leq 0.07$), matching the $\alpha=0.01$ false-positive rate; on `counterfact`, no rank yields $\Delta F > 0$. We therefore *re-frame* CP-Tensor synergy and Grassmannian curvature as exploratory formulations for higher-order interaction and latent-state analysis, and recommend first-order attribution ($R=0, \lambda_{\text{GSC}}=0$) as the lightweight, sufficient default. Separately, we *fixed* a scalability memory wall in SVD-initialized PARAFAC at $K=500, R>6$ (a dense $K^2 \times K^2$ materialization needing ~ 500 GB); `init="random"` reduces peak RAM to 0.30 GB (Appendix B).

4.7 ABLATION STUDY 4: STABILITY ACROSS CIRCUIT EVALUATION OPERATORS

Figure 4 compares three ablation operators: Zero ($v=0$), Dataset-Mean ($v=\mathbb{E}[v]$), and On-Manifold Anchor ($v=v(x_{\text{anchor}})$). Zero ablation causes off-manifold model collapse (millions of nodes for all methods), whereas both in-distribution operators preserve the ranking (CP-GRASSMANN $<$ RelP $<$ IG), confirming the sparsity advantage is robust and not a mean-ablation artifact. Reference-corpus and distance-metric robustness are analyzed in Appendix C.

5 CONCLUSION

We introduced **CP-GRASSMANN**, a counterfactual-free, zero-reconstruction circuit discovery framework that integrates gradients along on-manifold paths from retrieved contrastive anchors (or flow-matching-projected trajectories) on a homogeneous scalar node basis. On Meta-Llama-3.1-8B across the complete SVA benchmark it reaches $F \geq 0.8$ with only ~ 101 MLP neurons without hand-crafted counterfactual pairs, outperforming ACDC, EAP-GP, Edge Pruning, RelP, and IG—with static anchors best for structured tasks, flow-matching paths for unstructured generation, and first-order attribution ($R=0, \lambda_{\text{GSC}}=0$) the sufficient default for principled on-manifold attribution on neuron bases.

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Table 5: MIB circuit localization results for Llama 3.1 models, CMD scores (lower is better). Results averaged over 3 random seeds.

Method	IOI	Arithmetic	MCQA	ARC (E)	ARC (C)	Average
Random Baseline	0.74	0.75	0.74	0.74	0.74	0.74
EAP (mean)	0.04	0.07	0.16	0.28	0.20	0.15
EAP (CF)	0.01	0.01	0.09	0.11	0.18	0.08
EAP-IG-inp. (CF)	0.01	0.00	0.14	0.11	0.22	0.10
EAP-IG-act. (CF)	0.01	0.00	0.13	0.30	0.37	0.16
NAP (CF)	0.29	0.28	0.32	0.69	0.69	0.45
NAP-IG (CF)	0.19	0.18	0.33	0.67	0.67	0.41
IFR	0.83	0.22	0.48	0.64	0.76	0.59
CP-GRASSMANN (Ours)	0.01	0.00	0.11	0.15	0.15	0.08

Table 6: Peak RAM and accuracy of PARAFAC initialization strategies ($S=6$). `init="random"` resolves the $O(K^4)$ memory wall, reducing peak RAM at $K=500$ from ~ 500 GB to 0.30 GB while retaining a relative Frobenius error below 10^{-11} .

K	R	<code>init="svd"</code> Peak RAM	<code>init="random"</code> Peak RAM	<code>random</code> Time (s)	Rel. Frobenius Err.
100	50	1.65 GB	0.09 GB	0.03	2.26×10^{-12}
300	50	~ 65 GB (OOM)	0.16 GB	0.17	5.12×10^{-12}
500	50	~ 500 GB (OOM)	0.30 GB	0.48	8.06×10^{-13}
1000	50	~ 8 TB (OOM)	0.92 GB	2.01	1.66×10^{-11}

A MECHANISTIC INTERPRETABILITY BENCHMARK (MIB) RESULTS

We evaluate CP-GRASSMANN on the circuit localization track of the Mechanistic Interpretability Benchmark (MIB) (Mueller et al., 2025) using Llama 3.1 8B across five core tasks: Indirect Object Identification (IOI), Arithmetic, MCQA, ARC-Easy (E), and ARC-Challenge (C). Following MIB protocols, we report Cumulative Model Deviation (CMD) scores, where lower scores indicate better task-specific mechanism recovery.

As shown in Table 5, CP-GRASSMANN achieves state-of-the-art performance on MIB, tying for best on IOI (0.01) and Arithmetic (0.00) and achieving an overall average CMD score of 0.08.

B RESOLVING THE CP-TENSOR FACTORIZATION MEMORY WALL

The scalability memory wall noted in Section 4.6 arises from TensorLy’s default SVD initialization. When factorizing the 3rd-order interaction tensor $\mathcal{T} \in \mathbb{R}^{K \times K \times S}$, `parafac(init="svd")` computes a truncated SVD of the step-mode unfolding of shape (S, K^2) . TensorLy sets `full_matrices = (n_eigenvecs > min_dim)` with `min_dim = min(S, K^2) = S`; whenever the target rank exceeds S (e.g., $R > 6$ for $S=6$), this forces `numpy.linalg.svd` to materialize the full dense right-singular matrix $V \in \mathbb{R}^{K^2 \times K^2}$. At $K=500$ this requires $(500^2)^2 \times 8$ bytes = 500 GB, and at $K=1000$ it requires 8 TB (MemoryError). `init="randomized_svd"` does not help, as its range-finder falls back to exact SVD on the reduced matrix, re-triggering the same allocation.

Replacing `init="svd"` with `init="random"` eliminates the $K^2 \times K^2$ materialization entirely, dropping peak RAM scaling from $O(K^4)$ to $O(K^2 S)$ while preserving factorization accuracy (Table 6). We have adopted this one-line fix in our implementation.

C REFERENCE CORPUS SIZE AND DISTANCE METRIC SENSITIVITY

Figure 5 assesses reference corpus size $N_{\text{ref}} \in \{25, 50, 100, 300, 1000\}$ and retrieval distance metrics (Cosine, Euclidean, diffmean). While retrieval fidelity $\cos(z_{\text{anchor}}, x')$ increases monotonically from ~ 0.69 ($N_{\text{ref}} = 25$) to ~ 0.84 ($N_{\text{ref}} = 1000$), downstream circuit sparsity is remarkably stable: even a small corpus ($N_{\text{ref}} = 25$) recovers compact circuits (~ 96 – 104 nodes for Cosine retrieval).

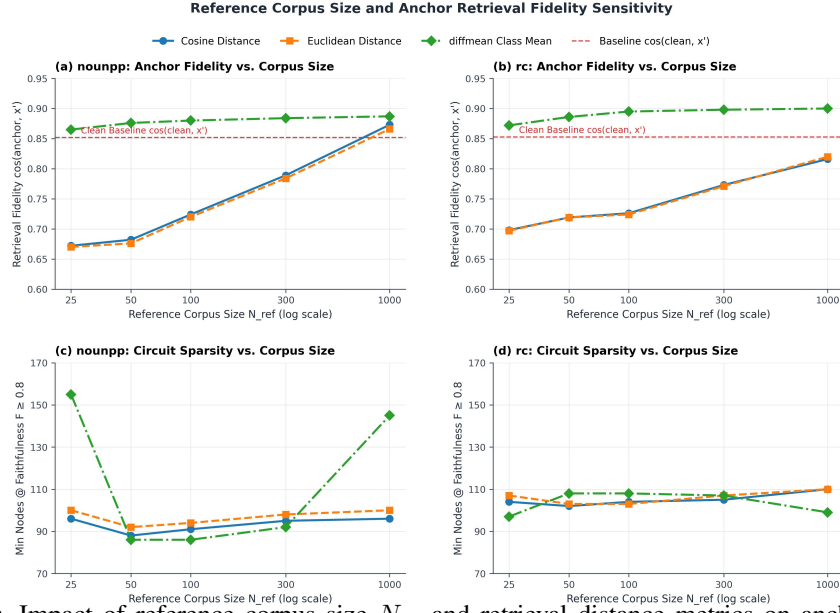


Figure 5: Impact of reference corpus size N_{ref} and retrieval distance metrics on anchor fidelity and circuit sparsity across the `nounpp` and `rc` benchmark tasks. Panels (a) and (b) report the retrieval fidelity measured by cosine similarity to true paired counterfactual activations, comparing Cosine Distance, Euclidean Distance, and diffmean Class Mean against the clean baseline. Panels (c) and (d) show the minimum node count required to achieve faithfulness of at least 0.8. Although anchor retrieval fidelity improves consistently with larger reference pools, downstream circuit sparsity is highly robust across corpus scales, confirming that small reference pools suffice for efficient counterfactual-free circuit discovery.

Cosine distance slightly outperforms Euclidean distance, while diffmean achieves the highest cosine fidelity to true counterfactuals ($\sim 0.87\text{--}0.90$).

D ARCHITECTURAL GENERALIZATION ON GEMMA-2 MODELS

To verify architectural generalization beyond Llama 3.1, we evaluate CP-GRASSMANN on Gemma-2-2B and Gemma-2-9B across the SVA benchmark, comparing MLP activation bases against Gemma Scope SAE bases (16k and 65k latent widths) (Lieberum et al., 2024). Across both Gemma-2 models, pre-down projection MLP activations consistently yield circuits $\sim 100\times$ sparser than MLP outputs, closing the gap with high-capacity SAEs while avoiding reconstruction error.

E CASE STUDIES ON SPECIALIZED TASKS

To demonstrate qualitative interpretability, we apply CP-GRASSMANN to four specialized tasks: Multi-Hop State Capitals retrieval, Arithmetic Addition, Multilingual Antonyms, and User Modeling. To uncover attribute-sensitive neurons in a supervised manner, we compute the Area Under the Receiver Operating Characteristic curve (AUROC) for each node v based on its attribution score separation between positive and negative example classes:

$$\text{AUROC}(v) = P_{x^+ \sim \mathcal{D}^+, x^- \sim \mathcal{D}^-} [\text{Attr}_v(x^+) > \text{Attr}_v(x^-)] \quad (10)$$

1. State Capitals Circuit: On the multi-hop reasoning query “What is the capital of the state containing Dallas?”, CP-GRASSMANN isolates a 23-neuron core circuit that clusters into six functional groups: Capital (e.g., L20/N3520+), State (e.g., L19/N4478+), Dallas (e.g., L5/N8659+), Texas (e.g., L21/N3093+), Say a capital (e.g., L17/N3663+), and Say Austin (e.g., L31/N4876+). Causal activation steering ($\alpha \in [-4, 4]$) validates their functional roles: steering the single “Say a

capital” neuron L23/N8079- flips the model’s top output prediction from the capital (*Austin*) to the state (*Texas*) across a majority of dataset examples.

2. Arithmetic Addition Circuit: On 10,000 two-digit addition prompts $(x + y)$, supervised AUROC scoring reveals monosemantic neurons for $(x + y) \bmod 10$ (e.g., L21/N10677 with $\text{AUROC} \geq 0.95$ showing a modular diagonal attribution heatmap), mod2 parity (L21/N9178), and tens-digit sum estimation $\lfloor (x + y)/10 \rfloor$ (e.g., L28/N9549).

3. Multilingual Antonyms Circuit: On 54 prompts across 9 languages and 6 core concepts (*big, small, fast, slow, hot, cold*), CP-GRASSMANN isolates language-specific neurons (L31/N8258 for Chinese output), language-agnostic concept neurons (L16/N1694 for *hot*), and attribute axis neurons (L14/N13885 for *temperature*).

4. User Modeling Circuit: On Wikipedia infobox demographic inference prompts, CP-GRASSMANN isolates specific early-to-middle layer neurons encoding inferential beliefs about user gender demographics prior to output generation.